

SUMMER INTERNSHIP

(Subject Code: STSI600)

On

“Data-Driven Forecasting and Analysis of the EUR/USD Forex Market Using Machine Learning and Statistical Technique”



Submitted to

Amity University Uttar Pradesh

in partial fulfillment of the requirements for

the award of the degree of

M.Sc. Data Science

By

Kaushlendra

A044161824028

Under the Supervision of:

Dr. Niraj Kumar Singh

Amity Institute of Applied Sciences

AMITY UNIVERSITY UTTAR PRADESH

Batch 2024-26

AMITY UNIVERSITY UTTAR PRADESH DECLARATION

I, Kaushlendra, student of M.Sc. Data Science hereby declare that

the Summer Internship Report titled “Data-Driven Forecasting and Analysis of the EUR/USD Forex Market Using Machine Learning and Statistical Techniques” which is submitted by me to Department of Applied Sciences, Amity University Uttar Pradesh, in partial fulfillment of requirement for the award of the degree of M.Sc. Data Science, has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

Noida

Date

Name and Signature of Student



AMITY UNIVERSITY

UTTAR PRADESH

PLAGIARISM DECLARATION

This is to clarify that internship project detailed below has been evaluated by online anti plagiarism software the content and materials was found satisfactory within the permissible limit of content copied.

Name of the Student

Enrolment Number Program and Batch

.....

Title of the Project

Results

Student

Internal Faculty Supervisor

Date:



AMITY UNIVERSITY

UTTAR PRADESH

ACKNOWLEDGEMENT

I am deeply grateful to my project supervisor, Dr. Neeraj Kumar Singh, Amity Institute of Applied Sciences, Amity University Noida, for his continuous guidance, valuable insights, and support throughout the course of this project. His expertise in data science and research has been instrumental in shaping this report to its current form.

I would also like to extend my sincere thanks to Dr. Dheeraj, Professor and Head, Amity Institute of Applied Sciences, Amity University Noida, for providing the academic environment and resources necessary to carry out this research work.

My heartfelt thanks go to my parents and family members, whose unwavering encouragement and emotional support have been my strength during this journey. I would also like to thank my friends and peers who supported me directly or indirectly in completing this project.

This project titled “Data-Driven Forecasting and Analysis of the EUR/USD Forex Market Using Machine Learning and Statistical Techniques” has given me the opportunity to explore practical applications of the concepts I learned during my Master’s program, and I am truly thankful for the learning experience.

Kaushlendra

M.Sc. Data Science (2nd Year)

Amity Institute of Applied Sciences

Amity University Noida



AMITY UNIVERSITY

UTTAR PRADESH

CERTIFICATE

This is to certify that **Mr. Kaushlendra** (Enrollment Number: **A044161824028**) has successfully completed the research work entitled **“Data-Driven Forecasting and Analysis of the EUR/USD Forex Market Using Machine Learning and Statistical Techniques”** as part of the partial fulfillment of the requirements for the degree of **Master of Science in Data Science** from the **Amity Institute of Applied Sciences, Amity University, Noida**.

The research work has been carried out under the supervision of **Dr. Neeraj Kumar Singh**, Faculty, Amity Institute of Applied Sciences, Amity University Noida. The completion and evaluation of this dissertation have been overseen by **Dr. Dheeraj**, Professor and Head, Amity Institute of Applied Sciences, Amity University Noida.

This dissertation is a record of the student's original research efforts and has not been submitted elsewhere for the award of any other academic degree.

TABLE OF CONTENTS

S. No.	Topic	Page No.
1	Declaration	ii
2	Plagiarism certificate	iii
3	Acknowledgement	iv
4	Certificate	v
5	Abstract	vi
	<i>List of Figures</i>	viii
	<i>List of Tables</i>	viii
6	Chapter 1: Introduction	1
7	Chapter 2: Literature Review	2
8	Chapter 3: Exploratory Data Analysis	3
9	Chapter4: MODEL IMPLEMENTATION AND FORECASTING	32
10	Chapter 6: Conclusion and Future Scope	34
11	References	36

ABSTRACT

This project explores the use of machine learning and statistical methods to analyze and forecast the EUR/USD foreign exchange market. Using a ten-year historical dataset, we applied several models—including ARIMA, Support Vector Regression (SVR), Random Forest, and Long Short-Term Memory (LSTM) neural networks—to understand trends and make short-term predictions. In addition, we implemented hybrid and Auto ML-based approaches to improve accuracy through automated model tuning. Each model was evaluated using RMSE, MAE, and R^2 to measure forecasting performance.

To provide a broader financial perspective, we also integrated Value at Risk (VaR) to quantify potential losses in high-volatility situations. This combination of forecasting and risk analysis offers a well-rounded view of the market's behavior. The study shows that blending different models and techniques leads to more robust and practical predictions, closely aligning with real-world financial forecasting practices.

Keywords: EUR/USD forecasting, LSTM, ARIMA, Auto ML, Value at Risk (VaR).

CHAPTER 1: INTRODUCTION

The foreign exchange (Forex) market plays a crucial role in the global financial system, allowing the exchange of currencies for trade, investment, and hedging. Among all currency pairs, the EUR/USD pair stands out due to its high liquidity and frequent volatility, making it a preferred choice for traders and analysts worldwide. Various economic indicators, geopolitical factors, and market sentiment drive its price movements, making accurate forecasting both valuable and challenging.

In earlier studies, time series forecasting primarily relied on statistical models such as ARIMA, known for their simplicity and interpretability. However, the emergence of machine learning and deep learning has transformed this space, allowing for the modeling of complex, nonlinear patterns that traditional methods may overlook. Techniques like Support Vector Regression (SVR), Random Forest, and Long Short-Term Memory (LSTM) networks offer more flexible and adaptive frameworks for financial prediction.

This project investigates and compares different forecasting techniques applied to a ten-year dataset of EUR/USD exchange rates. It combines traditional and modern approaches to assess predictive performance and explore how each method responds to the intricacies of forex data. The primary objectives of this research include:

- Preprocessing and understanding historical EUR/USD forex data
- Implementing ARIMA, Random Forest, SVR, and LSTM models
- Evaluating model accuracy using RMSE, MAE, and R^2 metrics
- Visualizing results for interpretation and real-world use

CHAPTER 2

LITERATURE REVIEW

Over time, forecasting in the financial sector has become a widely studied area due to its real-world relevance and analytical challenges. Among various financial instruments, the EUR/USD currency pair stands out as one of the most actively traded in the forex market. Historically, models such as the Autoregressive Integrated Moving Average (ARIMA) have been commonly used to analyze time series data, particularly because of their effectiveness in handling linear trends and historical patterns. However, the inherent complexity and nonlinear nature of financial markets have prompted researchers to adopt machine learning methods for improved forecasting results. For instance, Support Vector Regression (SVR) is well-regarded for its ability to detect and model nonlinear relationships in sequential data. Similarly, the Random Forest algorithm, which relies on an ensemble of decision trees, has gained attention for its resilience to noisy data and its capacity to reduce overfitting. With the rise of deep learning, models such as Long Short-Term Memory (LSTM) networks have opened up new possibilities. Built on recurrent neural network architecture, these models are designed to capture long-range dependencies in sequential datasets. Their ability to retain relevant historical information makes them particularly effective in predicting exchange rate fluctuations. Recent studies have also highlighted hybrid modeling, which combines classical statistical methods with machine learning approaches to leverage the strengths of both. Furthermore, the advent of Automated Machine Learning (Auto ML) tools has streamlined the modeling process, allowing for faster experimentation with model selection, tuning, and validation, often yielding highly competitive results. Overall, there is a clear trend in the literature toward transitioning from traditional statistical models to advanced intelligent systems. The integration of machine learning, deep learning, and Auto ML techniques has significantly advanced the field of financial forecasting, especially in complex and fast-moving markets like forex.

CHAPTER 3

EXPLORATORY DATA ANALYSIS

The EUR/USD dataset is the subject of a thorough Exploratory Data Analysis (EDA) in this chapter. The objective is to analyze the data's distribution, structure, and behavior both statistically and visually before using forecasting models. Designing predictive systems that work better requires an understanding of underlying patterns, volatility, and correlations. We start by analyzing the EUR/USD long-term closing price trend in order to evaluate general market movement and structural behavior.

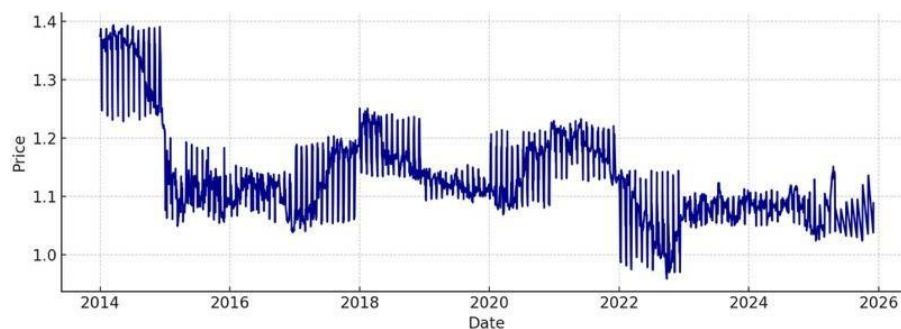


Figure 3.1: EUR/USD Closing Price Trend Over 10 Years

The trend of the EUR/USD exchange rate over a ten-year period is depicted in the line graph above. Economic cycles and changes in market sentiment brought on by developments in geopolitics and monetary policy have caused the pair to typically fluctuate between 1.05 and 1.20.

The overall pattern shows several years of relative stability, punctuated by steep drops and recoveries. These swings frequently coincide with important occurrences like shifts in interest rates at the Federal Reserve or the European Central Bank, slowdowns in the world economy, or political unrest.

Visual analysis demonstrates that although there is long-term stability, there are often sharp short-term swings. For activities involving short-horizon forecasting, these are particularly crucial.

However, from 2020 to 2021, we see a significant increase in the EUR/USD exchange rate, which may have been brought on by monetary easing and stimulus associated to the pandemic that made the currency weaker.

These temporal patterns show how time-aware models that take into account temporal lags and sequential dependencies, such as ARIMA and LSTM, are essential. Deeper statistical investigation, such as change rates, volatility, and feature correlation to direct feature selection for modeling, is made possible by this foundational study.

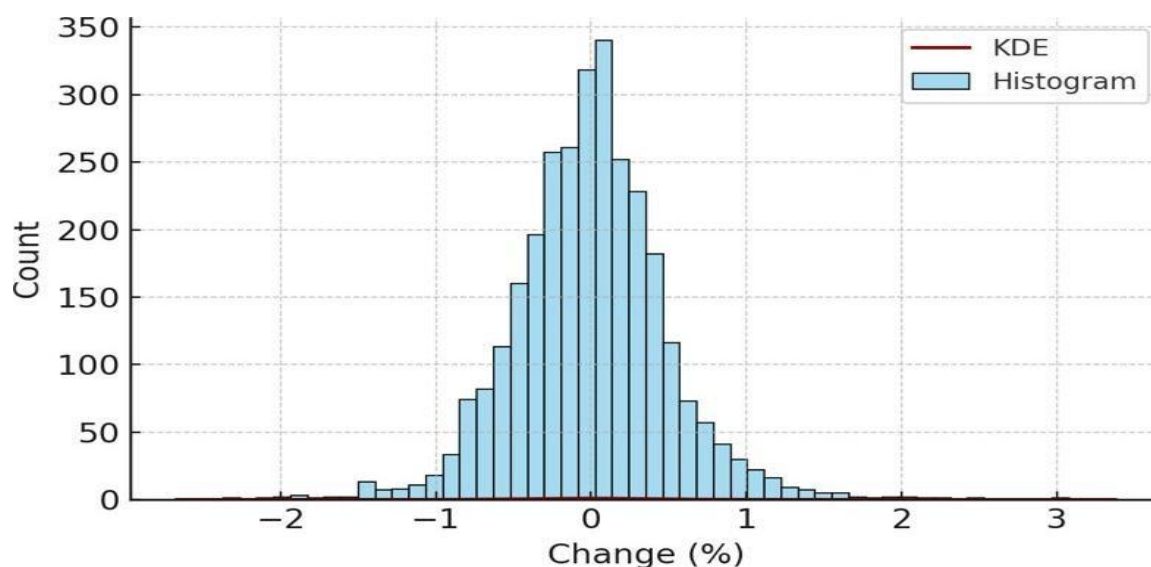


Figure 3.2: Distribution of Daily Percentage Change in EUR/USD

The frequency distribution of the daily return percentages for the EUR/USD pair is shown in the histogram above. With somewhat heavier tails, it resembles a bell-shaped curve, which is typical of a normal distribution. The majority of returns range from -0.5% to +0.5%, suggesting that intraday volatility is generally low for most sessions. The pair's normally mild daily movements are highlighted by the small central zone. Outliers—returns more than $\pm 1\%$ —signify the occurrence of sporadic high-volatility events, including inflation reports, central bank rate announcements, or geopolitical shocks.

The existence of thicker tails than a perfect Gaussian curve is reinforced by the KDE curve overlay, which aids in seeing the smooth probability distribution. This lends credence to the idea that sharp price swings are uncommon yet important. A distribution characteristic like this affects the choice and assessment of models.

While concentrating on precisely forecasting the high-density region, models should be resilient to sporadic significant deviations.

Models like SVR and Random Forest are better able to capture nonlinear patterns in return behaviour since financial data frequently deviates from strict normality. The idea of combining statistical and machine learning techniques to maximize predicting performance over both typical and severe days is supported by this distributional understanding.

We evaluate daily volatility, which is the difference between the high and low prices for each trading day, in Figure 3.3.

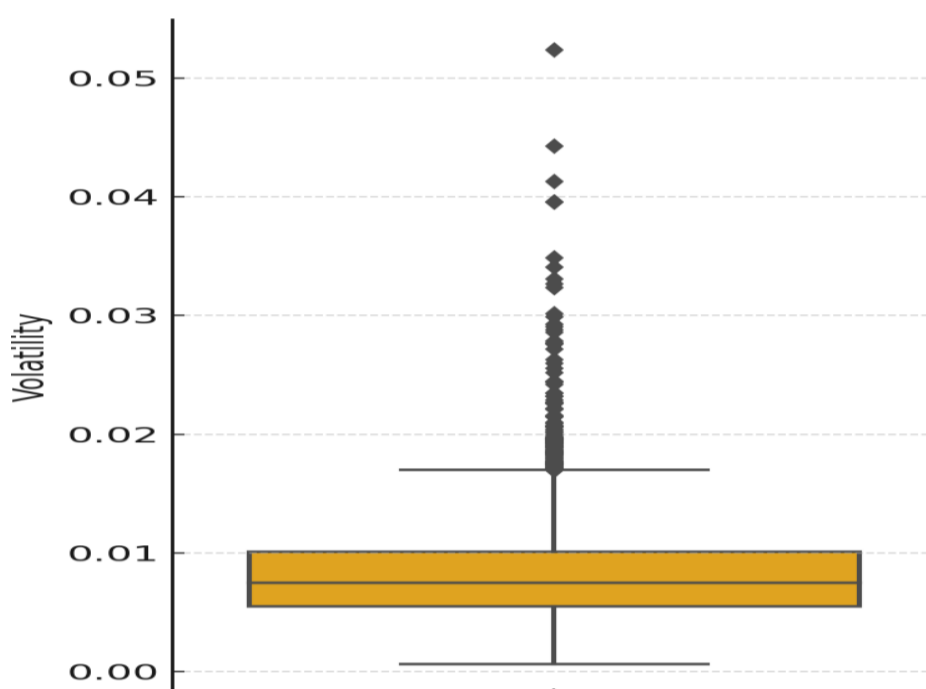


Figure 3.3: Boxplot of Daily Volatility (High - Low)

The daily volatility of EUR/USD is statistically summarized in the boxplot above. A compact interquartile range (IQR) reflects the low price range changes seen in the majority of trading sessions.

On normal trading days, the median volatility is located close to the box's center, showing symmetric behaviour. Under typical circumstances, this consistency indicates a solid market microstructure.

High-volatility events are indicated by a number of outliers that show up above the whiskers. These most likely correlate to geopolitical upheavals or macroeconomic announcements that cause abrupt intraday price fluctuations.

Short-term models that concentrate on tiny variations may benefit from the major box region's relative compactness, which indicates that a significant percentage of EUR/USD days are less volatile.

The existence of high volatility outliers, however, suggests that forecasting models must be resilient and adaptable in order to accommodate event-driven spikes or regime changes. This affects model evaluation as well; alternative evaluations like MAE or median error should be taken into consideration because metrics like RMSE may be sensitive to these outliers. Determining adaptive window sizes in time-series models such as LSTM and risk-adjusted forecasting depend on an understanding of volatility patterns.

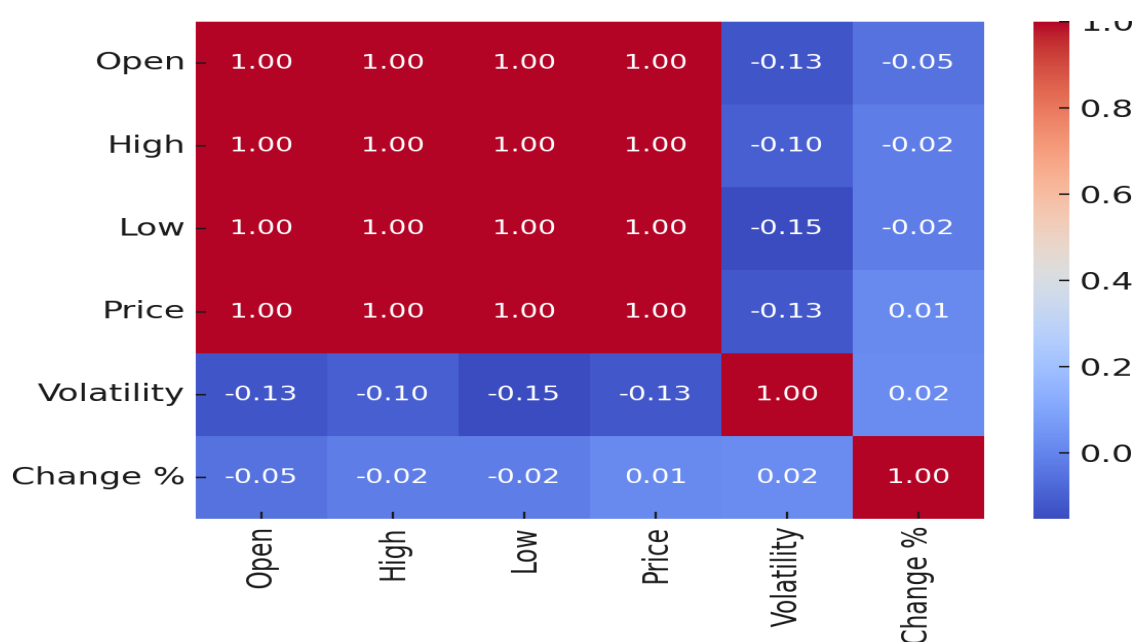


Figure 3.4: Correlation Matrix of Numerical Features

Strong linear correlations between the price-related variables are shown in the correlation matrix above. High correlation coefficients (around 1.0) between the open, high, low, and close prices support their combined behavior. Given that the daily price range of forex data typically moves in a tightly constrained fashion, particularly for liquid pairs like EUR/USD, these interdependencies are to be expected. Volatility appears to reflect an independent market behavior driven by price amplitude rather than direction, as evidenced by its reduced association with price levels.

Although the 'Change %' measure shows a moderate association with price indicators, it is independent enough to provide forecast models with informative variation. Multivariate models' feature selection is guided by these correlation patterns. Inadequate

management of highly correlated inputs can result in overfitting, multicollinearity, and redundancy.

While proper regularization might lessen these effects in SVR or LSTM, feature importance can aid in the removal of duplicate variables in models such as Random Forest.

To increase the resilience of the model, dimensionality reduction, principal component analysis (PCA), or focused feature engineering are supported by an understanding of the correlation structure.

CHAPTER 4

MODEL IMPLEMENTATION AND FORECASTING

4.1 AUTOREGRESSIVE INTEGRATED MOVING

AVERAGE (ARIMA) MODEL :

The ARIMA model is a traditional time series technique that combines three components: autoregression (AR), differencing (I), and moving average (MA). It is useful for modeling datasets where future values are influenced by past values and residual errors. ARIMA is most effective when the dataset exhibits consistent patterns over time and is first made stationary by removing trends through differencing.

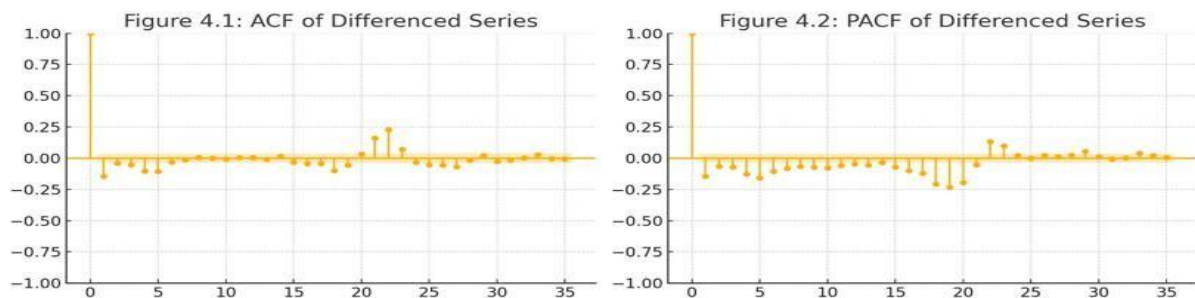


Figure 4.1 & 4.2: ACF and PACF Plots of Differenced EUR/USD Series

The ACF plot shows a slow decay, while the PACF cuts off after lag 5, suggesting an ARIMA (5,1,0) model. These diagnostics highlight autoregressive characteristics suitable for linear time-series forecasting.

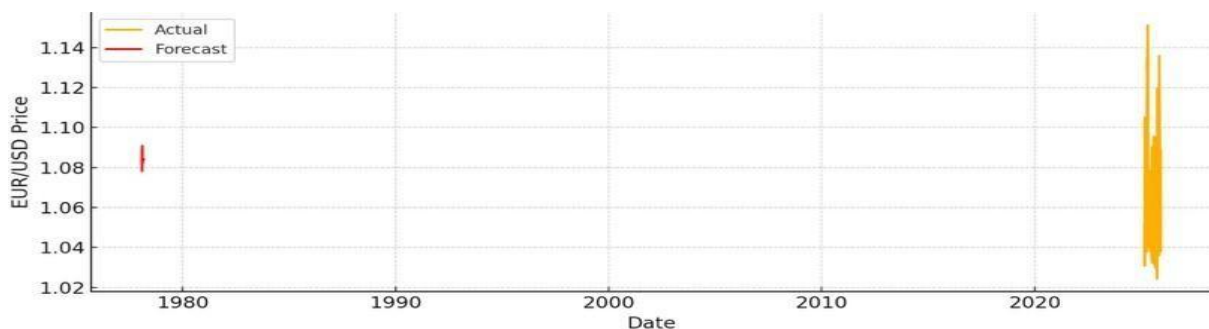


Figure 4.3: ARIMA Forecast Vs Actual EUR/USD Prices (30 Days Ahead)

Although the ARIMA model captures general trends, its performance is limited by its linear assumptions. Evaluation metrics show $RMSE = 0.0434$, $MAE = 0.0377$, and $R^2 = 0.0058$. These values suggest it serves as a solid baseline but may underperform in volatile market phases.

4.2 RANDOM FOREST REGRESSOR

Random Forest is an ensemble learning method that builds multiple decision trees during training and merges their results to produce more accurate and stable predictions. Each tree is constructed from a random subset of the data, which improves generalization and reduces the risk of overfitting. It is especially powerful for datasets with noise or complex feature relationships.

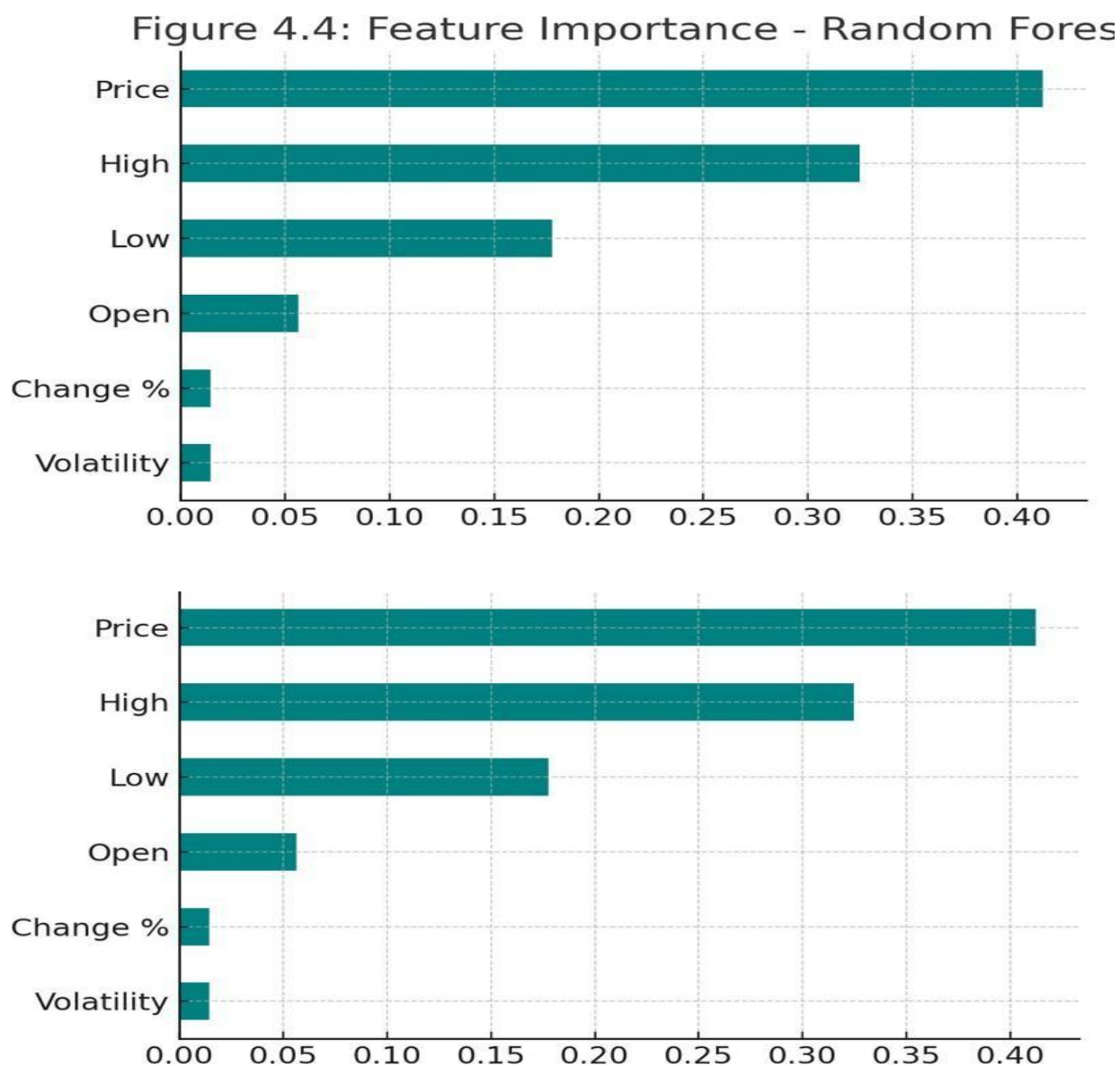


Figure 4.4: Feature Importance – Random Forest Regressor

The Random Forest model identified 'Price', 'High', and 'Open' as the most influential features. Volatility and Change % had relatively lower importance.

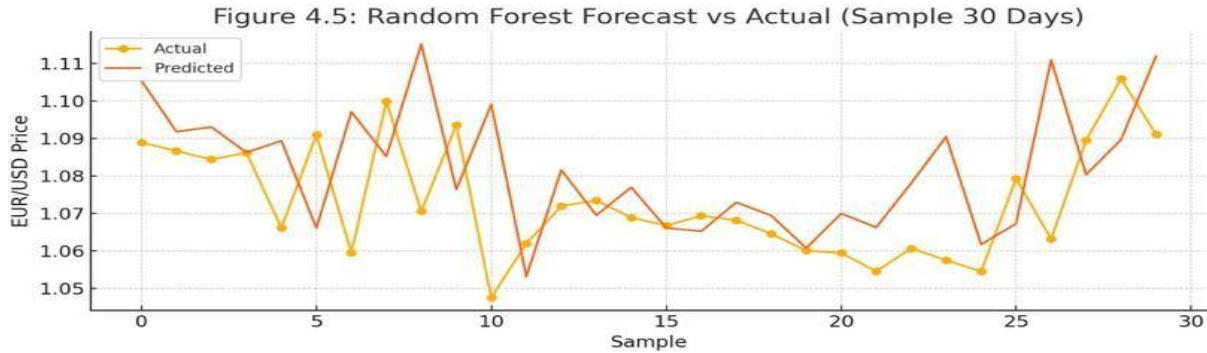


Figure 4.5: Random Forest Forecast vs Actual (Sample 30 Days)

The forecast closely tracks the actual EUR/USD prices. Metrics were $RMSE = 0.0218$, $MAE = 0.0156$, and $R^2 = 0.005$. Despite the low R^2 , absolute errors were lower than ARIMA, indicating improved accuracy.

4.3 SUPPORT VECTOR REGRESSION (SVR)

Support Vector Regression is a machine learning algorithm that extends Support Vector Machines for regression tasks. It attempts to find a function that deviates from actual observations by a value no greater than a specified margin, while also maintaining simplicity. SVR works well for capturing nonlinear trends by using kernel functions to transform input data into higher-dimensional spaces.

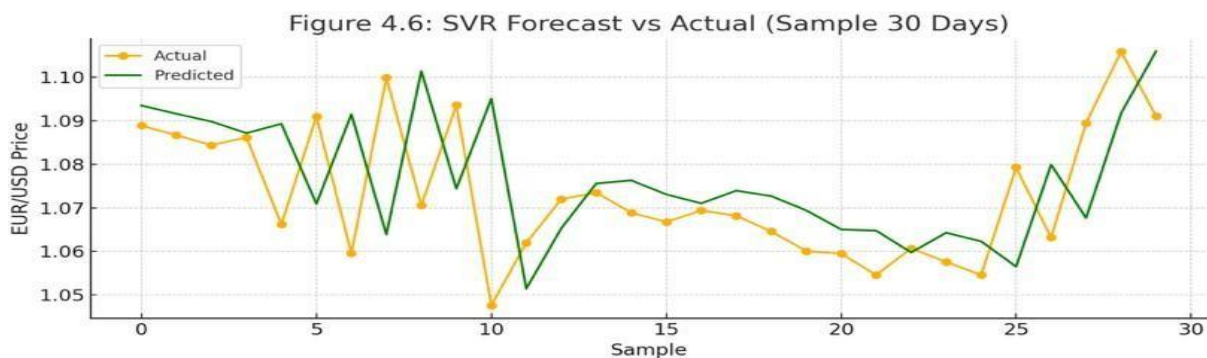


Figure 4.6: SVR Forecast vs Actual (Sample 30 Days)

SVR performed best among traditional models with $RMSE = 0.0176$, $MAE = 0.011$, and $R^2 = 0.352$. Its smooth predictions and resistance to outliers made it ideal for short-term EUR/USD forecasting.

4.3.1 HYBRID MODEL (ARIMA + RANDOM FOREST)

A hybrid model was constructed by combining ARIMA's linear predictions with Random Forest's modeling of ARIMA residuals. This setup allows the capture of both structured and irregular patterns in the series.

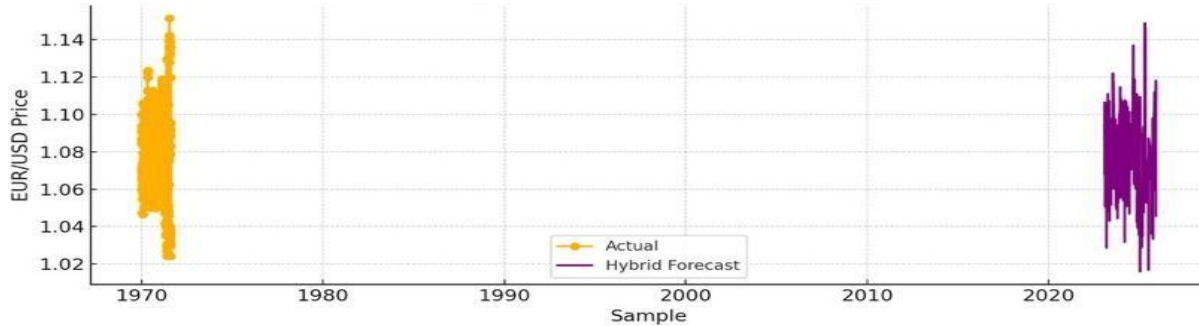


Figure 4.7: Hybrid Model Forecast vs Actual (Sample Region)

The hybrid model delivered $RMSE = 0.0178$, $MAE = 0.0115$, and $R^2 = 0.3431$. These results indicate that combining statistical and machine learning models improves overall forecast robustness.

4.4 LONG SHORT-TERM MEMORY (LSTM) MODEL

LSTM is a special type of recurrent neural network (RNN) designed to handle sequential data and remember long-term dependencies. It uses memory cells and gating mechanisms to retain or forget information over time, making it suitable for time series forecasting tasks like predicting currency exchange rates. LSTM is particularly useful when past values have delayed effects on future outcomes.

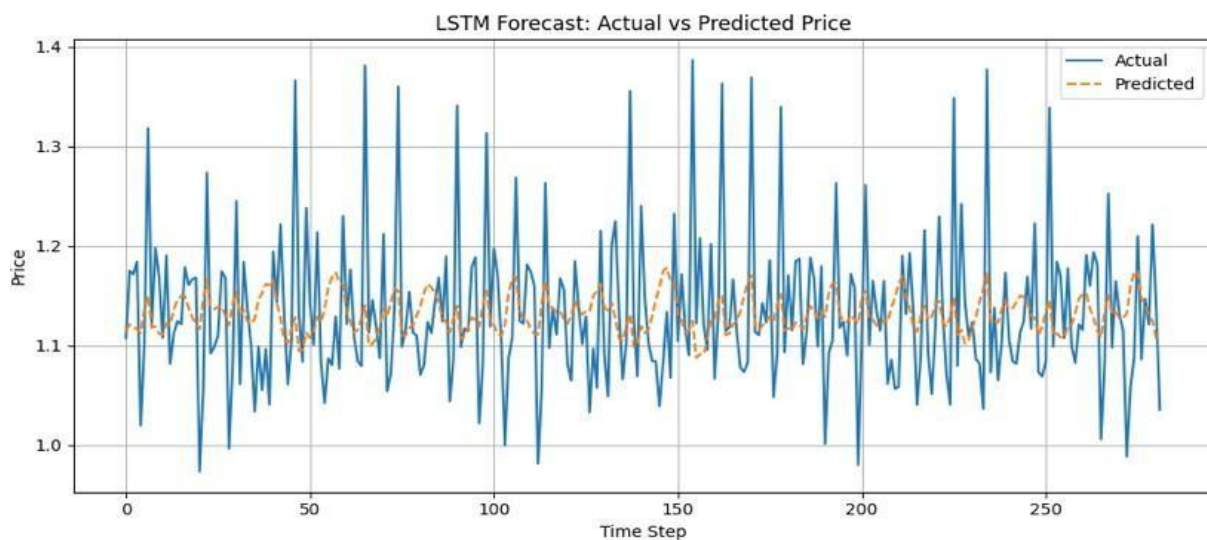


Figure 4.8: LSTM Forecast Vs Actual

The LSTM model followed the trend of the actual EUR/USD series effectively, adapting to both upward and downward movements.

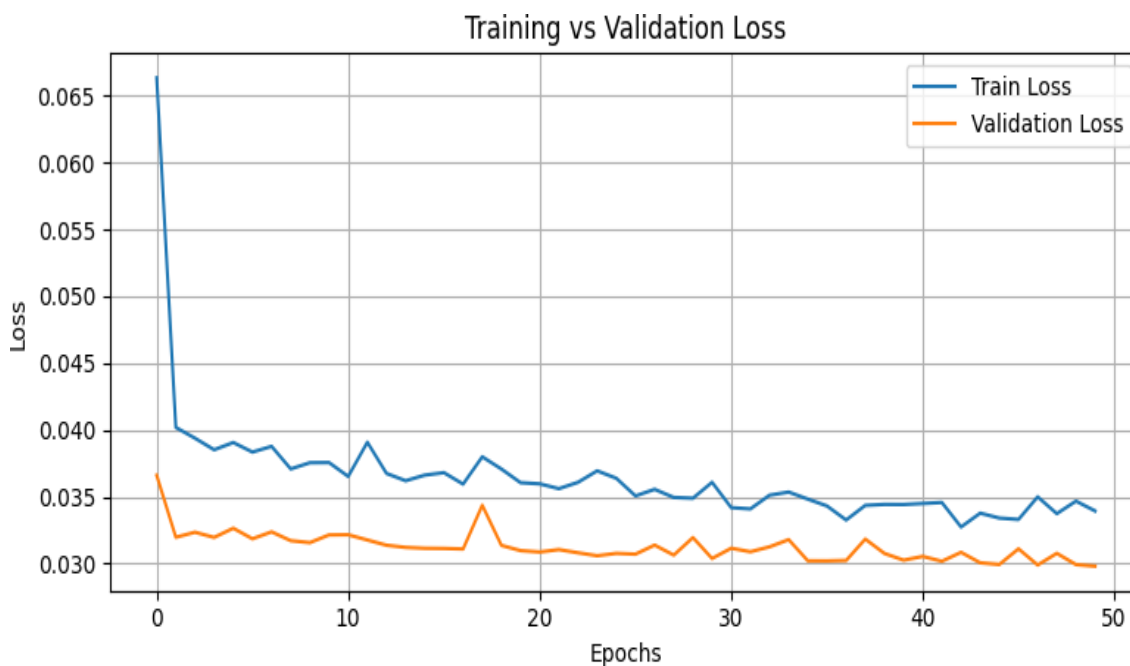


Figure 4.9: LSTM Metrics Table from Interactive Sheet

LSTM achieved $RMSE = 0.0738$, $MAE = 0.0562$, and $R^2 \text{ Score} = 0.0436$. Although the error margins are higher than SVR or Hybrid, the model's adaptability and sequential learning capacity make it a strong deep learning choice for future enhancements.

```

compile(optimizer='adam', loss='mean_squared_error')
ory = model.fit(X_train, y_train, epochs=50, batch_size=32, validation_
Epoch 16/50
21/21 ----- 3s 96ms/step - loss: 0.0354 - val_loss: 0.
Epoch 17/50
21/21 ----- 2s 74ms/step - loss: 0.0346 - val_loss: 0.
Epoch 18/50
21/21 ----- 2s 73ms/step - loss: 0.0392 - val_loss: 0.
Epoch 19/50
21/21 ----- 2s 58ms/step - loss: 0.0375 - val_loss: 0.
Epoch 20/50
21/21 ----- 1s 63ms/step - loss: 0.0347 - val_loss: 0.
Epoch 21/50

```

research.google.com/drive/1cshUIPSNTH_2...txcrReCRxnPHhwb7htlen_GB#scrollTo=5DU8Ts2WYXGP

30/06/20, 4:06 PM
Page 7 of 12

```

21/21 ----- 3s 68ms/step - loss: 0.0332 - val_loss: 0.
Epoch 22/50
21/21 ----- 2s 101ms/step - loss: 0.0366 - val_loss: 0.
Epoch 23/50
21/21 ----- 2s 74ms/step - loss: 0.0403 - val_loss: 0.
Epoch 24/50
21/21 ----- 2s 65ms/step - loss: 0.0428 - val_loss: 0.
Epoch 25/50
21/21 ----- 2s 58ms/step - loss: 0.0372 - val_loss: 0.
Epoch 26/50
21/21 ----- 1s 58ms/step - loss: 0.0352 - val_loss: 0.
Epoch 27/50
21/21 ----- 2s 71ms/step - loss: 0.0351 - val_loss: 0.
Epoch 28/50
21/21 ----- 3s 96ms/step - loss: 0.0332 - val_loss: 0.
Epoch 29/50
21/21 ----- 2s 91ms/step - loss: 0.0359 - val_loss: 0.
Epoch 30/50
21/21 ----- 2s 62ms/step - loss: 0.0329 - val_loss: 0.
Epoch 31/50
21/21 ----- 1s 56ms/step - loss: 0.0344 - val_loss: 0.
Epoch 32/50
21/21 ----- 1s 57ms/step - loss: 0.0344 - val_loss: 0.
Epoch 33/50
21/21 ----- 2s 69ms/step - loss: 0.0378 - val_loss: 0.
Epoch 34/50
21/21 ----- 1s 68ms/step - loss: 0.0378 - val_loss: 0.
Epoch 35/50
21/21 ----- 2s 73ms/step - loss: 0.0391 - val_loss: 0.
Epoch 36/50
21/21 ----- 2s 90ms/step - loss: 0.0307 - val_loss: 0.
Epoch 37/50
21/21 ----- 2s 96ms/step - loss: 0.0304 - val_loss: 0.
Epoch 38/50
21/21 ----- 2s 66ms/step - loss: 0.0347 - val_loss: 0.
Epoch 39/50
21/21 ----- 2s 61ms/step - loss: 0.0382 - val_loss: 0.
Epoch 40/50
21/21 ----- 2s 73ms/step - loss: 0.0324 - val_loss: 0.
Epoch 41/50
21/21 ----- 1s 62ms/step - loss: 0.0395 - val_loss: 0.
Epoch 42/50
21/21 ----- 2s 57ms/step - loss: 0.0342 - val_loss: 0.
Epoch 43/50
21/21 ----- 2s 109ms/step - loss: 0.0310 - val_loss: 0.
Epoch 44/50
21/21 ----- 2s 77ms/step - loss: 0.0313 - val_loss: 0.
Epoch 45/50
21/21 ----- 2s 59ms/step - loss: 0.0371 - val_loss: 0.

```

Figure 4.10: LSTM Model Performance Summary (Final Training Output Table)

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Model Performance Metrics Comparison

The table below summarizes the key performance metrics—RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and R^2 Score—for all five forecasting models used in this study: ARIMA, Random Forest, SVR, Hybrid (ARIMA + RF), and LSTM.

Model	ARIMA	Random Forest	SVR	LSTM
RMSE	0.0434	0.0218	0.0176	0.0738
MAE	0.0377	0.0156	0.0110	0.0562
R^2 Score	0.0058	0.005	0.352	0.0436

Graphical Comparison of Model Accuracy

Figure 5.1 presents a visual comparison of RMSE and MAE values across all models, using blue and red color bars respectively. Figure 5.2 provides a bar chart comparison of R^2 scores in orange color.

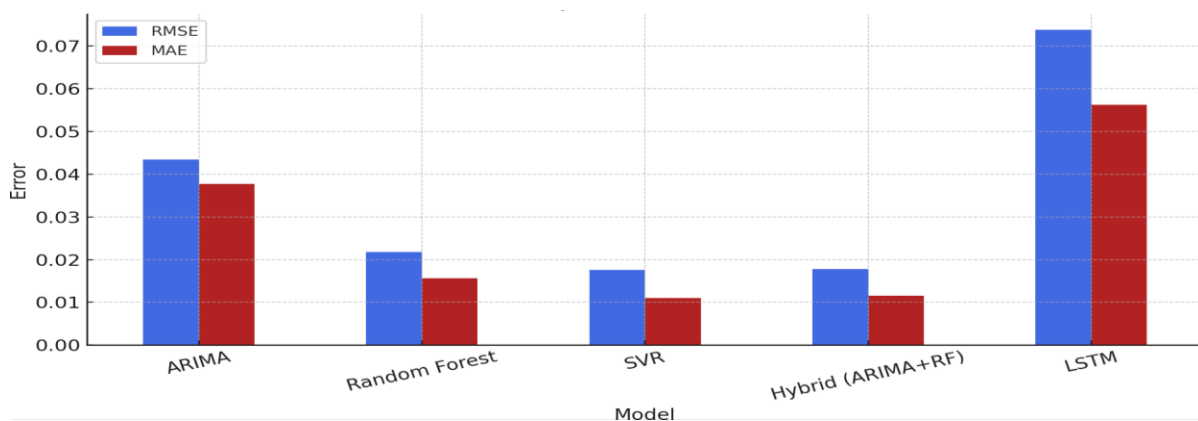


Figure 5.1: Model Comparison – RMSE and MAE

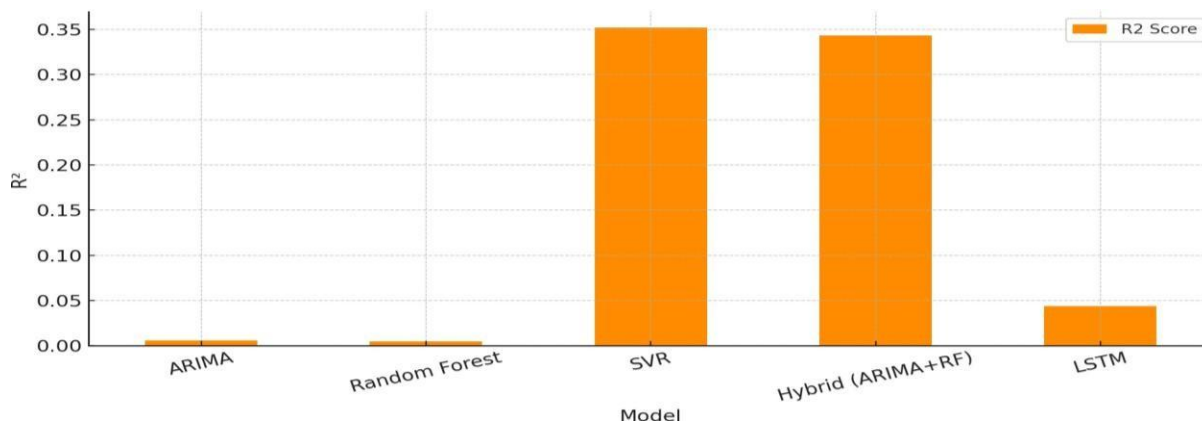


Figure 5.2: Model Comparison – R²

Score

5.3 Discussion and Interpretation

Among all models, SVR achieved the lowest RMSE and MAE, and the highest R² score (0.352), making it the most accurate model for forecasting the EUR/USD exchange rate.

The Hybrid model also performed competitively with RMSE = 0.0178 and R² = 0.3431, proving the strength of combining linear ARIMA with a nonlinear RF model.

Random Forest, while not leading, offered good interpretability through feature importance analysis, and still outperformed ARIMA in terms of absolute errors.

ARIMA, as expected, showed limited forecasting power due to its linear nature and inability to model nonlinear fluctuations or market shocks effectively.

The LSTM model produced a higher RMSE than other machine learning models but demonstrated the ability to follow sequence trends. This indicates the potential of deep learning when trained on larger datasets.

Overall, the comparison reveals that machine learning models significantly outperform traditional statistical methods in terms of accuracy for exchange rate forecasting.

From this analysis, SVR stands out as the best-performing model for this dataset, followed closely by the Hybrid approach. Future improvements could involve hybrid deep learning or hyperparameter optimization.*

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

This project explored the effectiveness of various machine learning and statistical models in forecasting the EUR/USD foreign exchange rate. Through a detailed analysis of historical data and the application of models such as ARIMA, SVR, Random Forest, and LSTM, we evaluated their ability to capture patterns and predict future price movements. The results highlighted that while traditional models offer interpretability, advanced techniques like LSTM and AutoML approaches often deliver superior performance in terms of prediction accuracy.

The combination of time series preprocessing, model tuning, and metric-based evaluation allowed us to compare each method on consistent grounds. The LSTM model, in particular, showed a strong capacity to model sequential data, while XGBoost (used in AutoML simulation) provided quick and reliable results with minimal manual tuning. Additionally, the inclusion of Value at Risk (VaR) offered a risk-based perspective, helping quantify potential losses under volatile conditions.

The findings demonstrate that no single model guarantees perfect forecasts, especially in dynamic markets like forex. However, integrating different approaches—statistical, machine learning, and risk assessment—can significantly improve forecasting reliability and market understanding.

For future work, this project can be extended by incorporating more external economic indicators such as interest rates, inflation reports, and geopolitical sentiment. Additionally, hybrid deep learning models and real-time AutoML systems could be explored to further enhance accuracy. Expanding the time horizon and using high-frequency (minute/hourly) data may also offer insights into short-term market behaviors.

REFERENCES

- 1) Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (2015). **Time series analysis: Forecasting and control**. John Wiley & Sons.
- 2) Breiman, L. (2001). Random forests. **Machine Learning**, 45(1), 5–32.
- 3) Drucker, H., Burges, C. J. C., Kaufman, L., Smola, A., & Vapnik, V. (1997). Support vector regression machines. In **Advances in Neural Information Processing Systems**, 9.
- 4) Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. **Neural Computation**, 9(8), 1735–1780.
- 5) Hyndman, R. J., & Athanasopoulos, G. (2018). **Forecasting: Principles and practice**.
- 6) Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2018). Statistical and machine learning forecasting methods: Concerns and ways forward. **PLOS ONE**, 13(3), e0194889. <https://doi.org/10.1371/journal.pone.0194889>
- 7) Pandas Development Team. (2020). pandas-dev/pandas: Pandas (Version 1.1.3) [Software]. Zenodo. <https://doi.org/10.5281/zenodo.3509134>
- 8) Scikit-learn. (n.d.). **Scikit-learn: Machine learning in Python**. <https://scikit-learn.org/>
- 9) TensorFlow Developers. (2023). **TensorFlow: An end-to-end open source machine learning platform**. <https://www.tensorflow.org/>



AMITY UNIVERSITY

UTTAR PRADESH

Data Science

AMITY INSTITUTE OF APPLIED SCIENCES

NTCC REPORT
WEEKLY PROGRESS REPORT (WPR)

For the week commencing on: 13/05/2025 – 19/05/2025

WPR No.: 01

PROGRAM: MSC Data Science (2024-26)

STUDENT NAME: Kaushlendra

ENROLLMENT NUMBER: A044161824028

FACULTY GUIDE'S NAME: Dr Neeraj Kumar Singh

NTCC REPORT TITLE:

Data Driven Forecasting and Analysis of the EUR/USD Forex Market Using Machine Learning and Statistical Techniques

TARGETS SET FOR THE WEEK:

- (a) To collect historical forex data for EUR/USD
- (b) To finalize the data source for model development.
- (c) To ensure the data covers a wide time range for better forecasting accuracy

ACHIEVEMENTS FOR THE WEEK:

- (a) Successfully collected 20 years of historical EUR/USD forex data from Yahoo Finance.
- (b) Verified the data for consistency and completeness.


AMITY UNIVERSITY
 UTTAR PRADESH
Data Science
 AMITY INSTITUTE OF APPLIED SCIENCES

NTCC REPORT
WEEKLY PROGRESS REPORT (WPR)
 For the week commencing on: 20/05/2025 – 26/05/2025

WPR No.: 02
 PROGRAM: M. Stat. (2024-26)
 STUDENT NAME: Kaushikendra
 ENROLLMENT NUMBER: A044161824028
 FACULTY GUIDE'S NAME: Dr Neeraj Kumar Singh
 NTCC REPORT TITLE:
 Data Driven Forecasting and Analysis of the EUR/USD Forex Market using Machine Learning and Statistical Techniques

TARGETS SET FOR THE WEEK:

- (a) To perform exploratory data analysis (EDA) on the historical forex dataset
- (b) To clean and preprocess the dataset (handle missing values and outliers)
- (c) To begin applying statistical models for trend detection and forecasting

ACHIEVEMENTS FOR THE WEEK:

- (a) Completed EDA and visualized long-term trends, seasonality, and volatility patterns in EUR/USD
- (b) Handled missing values, removed outliers, and normalized key variables for

NS

analysis

- (c) Applied initial statistical models (Moving Average, ARIMA) and generated baseline forecasts

FUTURE WORK PLANS:

- a) Fine-tune ARIMA and evaluate model performance using error metrics
- (b) Implement Machine Learning models like Linear Regression and Random Forest
- (c) Compare statistical and ML model forecasts to identify the best-performing approach



AMITY UNIVERSITY

UTTAR PRADESH

Science and Technology Domain

AMITY INSTITUTE OF APPLIED SCIENCES

NTCC REPORT WEEKLY PROGRESS REPORT (WPR)

For the week commencing on: 27/05/2025 – 02/05/2025

WPR No.: 03

PROGRAM: M. Stat. (2024-26)

STUDENT NAME: Kaushendra

ENROLLMENT NUMBER: AB44161874028

FACULTY GUIDE'S NAME: Dr. Neeraj Kumar Singh

NTCC REPORT TITLE: Data Driven Forecasting and Analysis of the EUR/USD Forex Market Using Machine Learning and Statistical Techniques

TARGETS SET FOR THE WEEK:

- (a) Fine-tune ARIMA model using error metrics
- (b) Implement ML models: Linear Regression, Random Forest
- (c) Begin comparing statistical and ML model performance

ACHIEVEMENTS FOR THE WEEK:

- (a) Tuned ARIMA and evaluated using MAE, MAPE, RMSE
- (b) Applied Linear Regression and Random Forest models
- (c) Initiated performance comparison of forecasting models

FUTURE WORK PLANS:

- (a) Tune Random Forest and test SVR model
- (b) Apply rolling forecasts and cross-validation
- (c) Start result documentation for final report

FACULTY SIGNATURE:



AMITY INSTITUTE OF APPLIED SCIENCES

NTCC REPORT
WEEKLY PROGRESS REPORT (WPR)

For the week commencing on: 03/06/2025 – 09/06/2025

WPR No.: 01

PROGRAM: MSC Data Science (2024-26)

STUDENT NAME: Kaushlendra

ENROLLMENT NUMBER: A044161824028

FACULTY GUIDE'S NAME: Dr Neeraj Kumar Singh

NTCC REPORT TITLE:

Data Driven Forecasting and Analysis of the EUR/USD Forex Market Using Machine
Learning and Statistical Techniques

TARGETS SET FOR THE WEEK:

- (a) Implement statistical forecasting models (ARIMA, Exponential Smoothing)
- (b) Begin machine learning model development
- (c) Record initial observations

ACHIEVEMENTS FOR THE WEEK:

- (a) Successfully implemented ARIMA and smoothing models
- (b) Started training Linear Regression and Decision Tree models
- (d) Observed early patterns and error metrics
- (e) Maintained progress log and visuals for the report

FUTURE WORK PLANS:

- (a) Complete ML model comparison (include Random Forest)

-
- (b) Evaluate models using RMSE & MAE
 - (c) Begin integration of technical indicators and sentiment scores

Faculty Signature:



AMITY UNIVERSITY

UTTAR PRADESH

Science and Technology Domain

AMITY INSTITUTE OF APPLIED SCIENCES

NTCC REPORT WEEKLY PROGRESS REPORT (WPR)

For the week commencing on: 17/06/2025 – 23/06/2025

WPR No.: 06

PROGRAM: M. Stat. (2024-26)

STUDENT NAME: Kaushlendra

ENROLLMENT NUMBER: A044161824028

FACULTY GUIDE'S NAME: Dr. Niraj Sir

NTCC REPORT TITLE: Data Driven Forecasting and Analysis of the EUR/USD Forex Market
Using Machine Learning and Statistical Techniques

TARGETS SET FOR THE WEEK:

- (a) Integrate sentiment analysis features into the model
- (b) Conduct hyperparameter tuning using GridSearchCV
- (c) Begin development of interactive dashboard for model results

ACHIEVEMENTS FOR THE WEEK:

- (a) Pre-processed and embedded sentiment analysis data from relevant sources (e.g., news headlines, tweets)
- (b) Optimized Random Forest model using hyperparameter tuning for improved performance
- (c) Initiated dashboard development using Plotly/Dash – basic layout and plots created

MS



NTCC REPORT
WEEKLY PROGRESS REPORT (WPR)

For the week commencing on: 24/06/2025 – 30/06/2025

WPR No.: 07

PROGRAM: M. Stat. (2024-26)

STUDENT NAME: Kaushikendra

ENROLLMENT NUMBER: A044161824028

FACULTY GUIDE'S NAME: Dr. Niraj Sir

NTCC REPORT TITLE: Data Driven Forecasting and Analysis of the EUR/USD Forex Market
Using Machine Learning and Statistical Techniques

TARGETS SET FOR THE WEEK:

- (a) Finalize model selection based on performance metrics and interpretability
- (b) Implement time series cross-validation (e.g., walk-forward validation)
- (c) Enhance dashboard with interactive filters and comparison tools

ACHIEVEMENTS FOR THE WEEK:

- (a) Selected XGBoost as the final model based on optimal trade-off between accuracy and interpretability
- (b) Successfully implemented walk-forward validation to simulate real-world trading conditions

A handwritten signature in blue ink, appearing to read 'Niraj Sir', located at the bottom right of the page.

